GAD vs Sella on HIP — 3-method benchmark (reduced)

2026-04-28 reduced. Transition1x train, n=300, 6 noise levels, 2000 steps, $f_{\max}<0.01 \wedge n_{\text{neg}}=1$, IRC = `sella_hip`.

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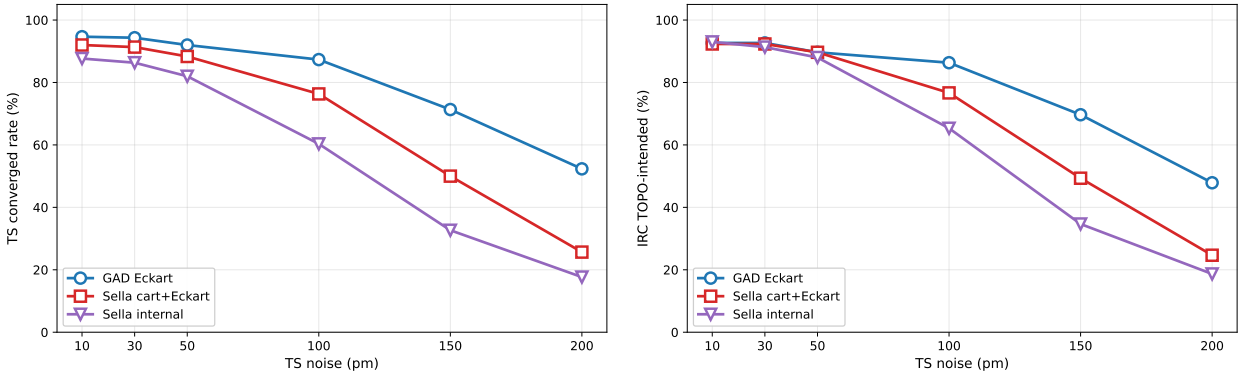
1 Headline



(a) TS-converged rate vs. noise.

(b) IRC TOPO-intended rate vs. noise.

Figure 1: Three canonical methods: GAD Eckart, Sella cart+Eckart, Sella internal.

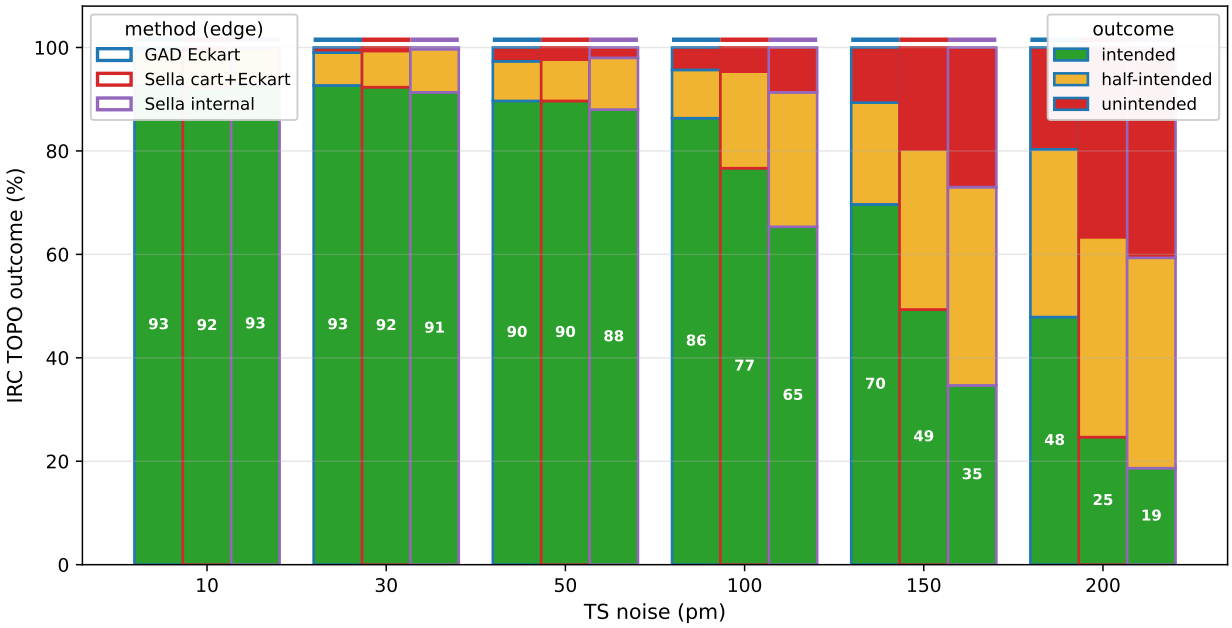


Figure 2: IRC TOPO outcome breakdown per method per noise. Bar fill = outcome (intended / half / unintended). Bar edge color = method.

TS-converged rate (%, n=300)

Method	10	30	50	100	150	200
GAD Eckart	87.3	87.3	85.3	**80.0**	**63.7**	**45.3**
Sella cart+Eckart	**92.0**	**91.3**	**88.3**	76.3	50.0	25.7
Sella internal	87.7	86.3	82.0	60.3	32.7	17.0

IRC TOPO-intended (%, n=300)

Method	10	30	50	100	150	200
GAD Eckart	**93.0**	**93.0**	**90.7**	**86.7**	**70.0**	**47.0**
Sella cart+Eckart	92.3	92.3	89.7	76.7	49.3	24.7
Sella internal	**93.0**	91.3	88.0	65.3	34.7	18.7

2 GAD step-size over time

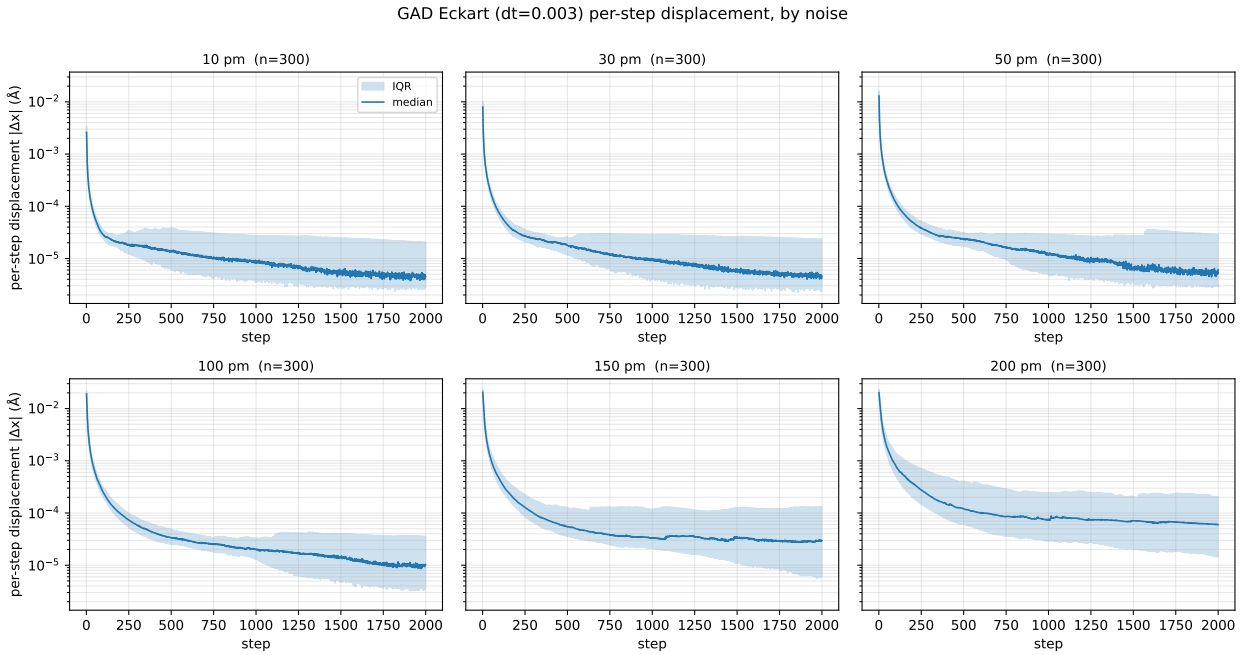
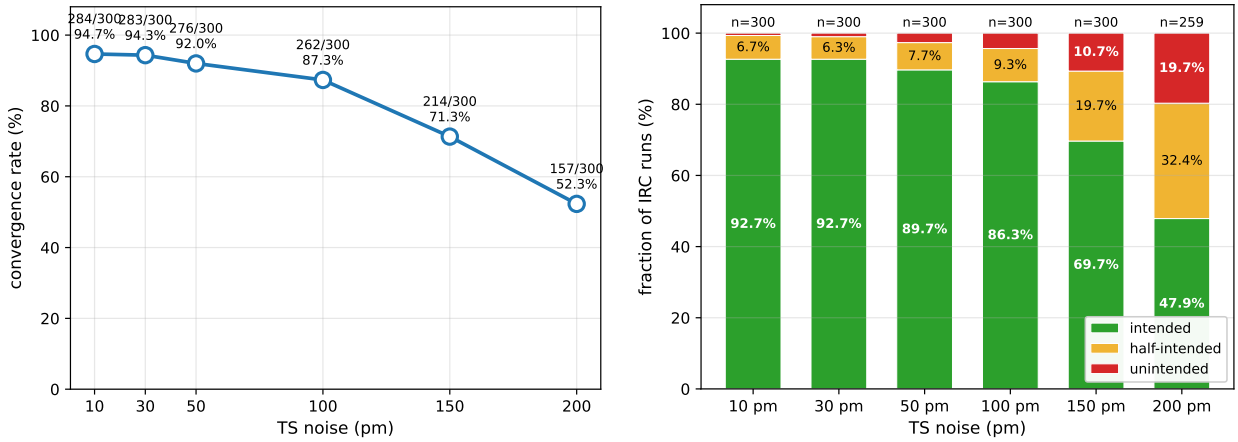


Figure 3: GAD Eckart (dt=0.003) per-step Cartesian displacement. Median (line) and IQR (band), aggregated over 300 trajectories per noise.

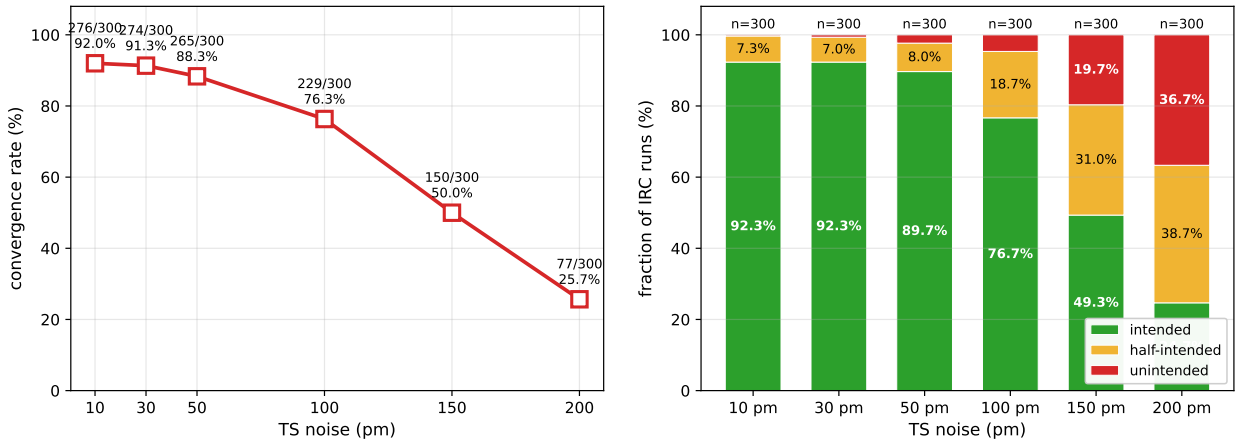
3 Per-method breakdown



(a) GAD Eckart — TS converged

(b) GAD Eckart — IRC TOPO

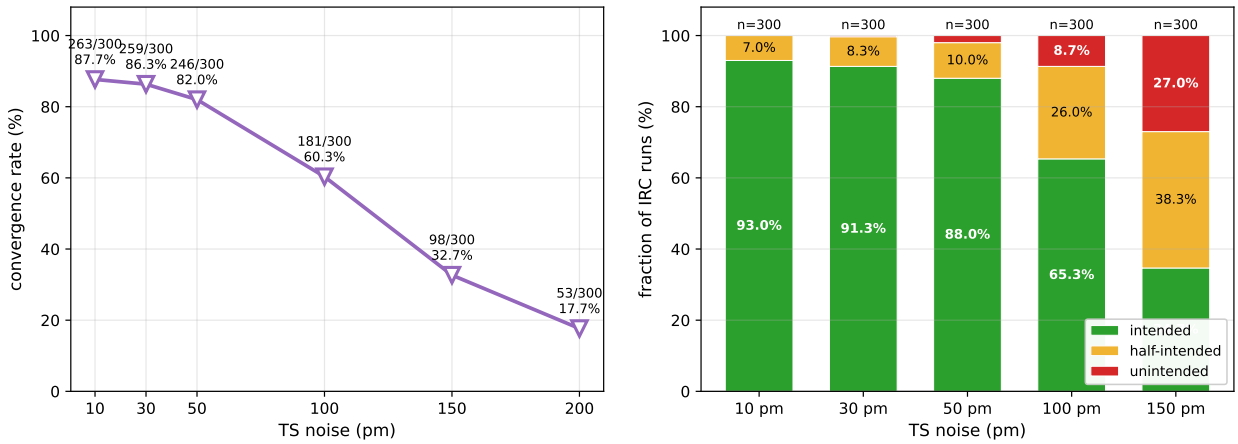
Figure 4: `gad_dt003` with Eckart projection, dt=0.003, 2000 steps.



(a) Sella cart+Eckart — TS converged

(b) Sella cart+Eckart — IRC TOPO

Figure 5: Sella cartesian + Eckart, 2000 steps, HIP Hessian every step.

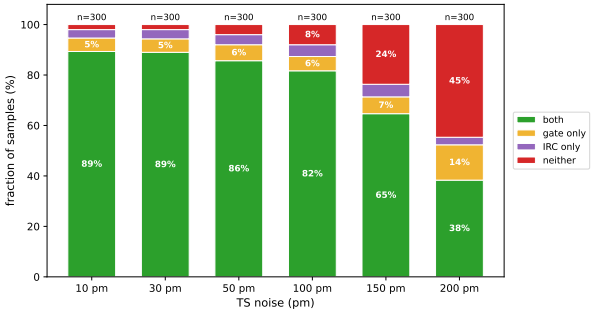


(a) Sella internal — TS converged

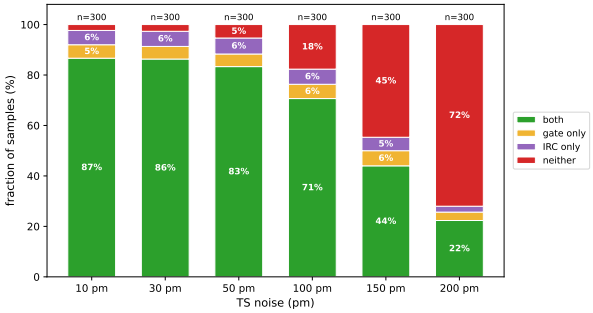
(b) Sella internal — IRC TOPO

Figure 6: Sella internal coords, 2000 steps, HIP Hessian every step.

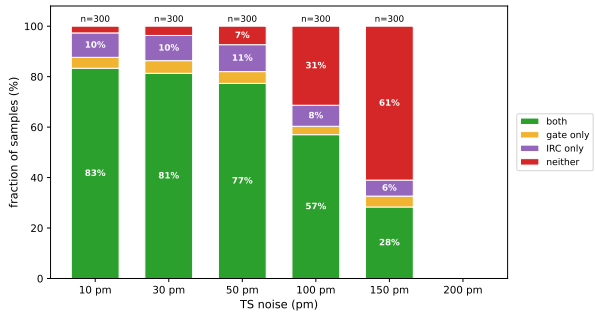
4 Gate vs IRC sample overlap



(a) GAD Eckart



(b) Sella cart+Eckart



(c) Sella internal

Figure 7: Per-method TS-converged vs. IRC-TOPO sample overlap. Green = both pass. Yellow = converged only. Purple = IRC only. Red = neither.

5 Data locations

Artifact	Path
GAD Eckart summaries	`runs/round2/`, `round3/`
Sella cart+Eckart summaries	`runs/sella_2000/`
Sella internal summaries	`runs/sella_2000/`
GAD IRC	`runs/irc_sellahip_allendpoints/`
Sella cart+Eckart IRC	`runs/irc_sella_carte_2000/`
Sella internal IRC	`runs/irc_sella_int_2000/`
This report	`IRC_COMPREHENSIVE_2026-04-28-reduced.tex/pdf`